

RESEARCH ARTICLE

Clustering and unconstrained ordination with Dirichlet process mixture models

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Funding information

U.S. Geological Survey under Grant/
Cooperative Agreement, Grant/Award
Number: G23AC00118

Handling Editor: Aldo Compagnoni

Abstract

1. Assessment of similarity in species composition or abundance across sampled locations is a common goal in multi-species monitoring programs. Existing ordination techniques provide a framework for clustering sample locations based on species composition by projecting high-dimensional community data into a low-dimensional, latent ecological gradient representing species composition. However, these techniques require specification of the number of distinct ecological communities present in the latent space, which can be difficult to determine in advance.
2. We develop an ordination model capable of simultaneous clustering and ordination that allows for estimation of the number of clusters present in the latent ecological gradient. This model draws latent coordinates for each sample location from a Dirichlet process mixture model, affording researchers with probabilistic statements about the number of clusters present in the latent ecological gradient.
3. The model is compared to existing methods for simultaneous clustering and ordination via simulation and applied to two empirical datasets; JAGS code to fit the proposed model is provided in an appendix. The first dataset concerns presence-absence records of fish in the Doubs river in eastern France and the second dataset describes presence-absence records of plant species in Craters of the Moon National Monument and Preserve (CRMO) in Idaho, USA. Results from both analyses align with existing ecological gradients at each location.
4. Development of the Dirichlet process ordination model provides wildlife managers with data-driven inferences about the number of distinct communities present across monitored locations, allowing for more cost-effective monitoring and reliable decision-making for conservation management.

KEYWORDS

clustering, dimension reduction, Dirichlet process mixture models, hierarchical Bayesian models, latent variable models, ordination

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1 | INTRODUCTION

Long-term monitoring programs have recently turned their focus from historical single-species monitoring toward monitoring entire species assemblages (Choi et al., 2019; Damgaard et al., 2020; Inoue et al., 2017; Zurell et al., 2020). To better understand the impacts of climate change, disease, wildfires and other emerging threats, many of these monitoring programs seek to assess similarity in species composition among sample locations across the monitored landscape or assess change in species composition over time (Jean et al., 2014; Nicolli, 2019; Stratton et al., 2019). To make these assessments, multivariate abundance data are typically collected at multiple locations within the monitored region. These abundance data can then be used to identify trends in species composition or cluster sample locations based on species composition. There exist many different techniques for making inferences about high-dimensional community data, including generalised linear mixed models, species distribution models and ordination techniques (Borcard et al., 2011; Pollock et al., 2014; Popovic et al., 2022; Roberts, 2020; van der Veen et al., 2021). This research focuses on methods for clustering and unconstrained ordination as tools for assessing similarity in species composition among sample locations as it allows researchers to cluster sample locations based on species composition and visualise results in a low-dimensional space, does not require environmental covariates and is widely implemented by ecologists and wildlife managers (Hui et al., 2015).

Algorithmic projection and clustering techniques may be combined in a two-step approach to conduct unconstrained ordination and cluster the results (Kéry & Royle, 2021, ch. 8). First, traditional unconstrained ordination techniques rely on constructing a pairwise dissimilarity matrix between sites based on some distance measure, for example, Bray–Curtis dissimilarity (Bray & Curtis, 1957), then projecting that matrix into a lower-dimensional space meant to represent the underlying dominant ecological gradient using some algorithm-based projection technique (ter Braak & Prentice, 1988). Common choices of projection techniques include classical multidimensional scaling (MDS, Kruskal & Wish, 1978) or non-metric multidimensional scaling (NMDS, Kruskal, 1964). In the second step, algorithmic clustering techniques are applied in the lower-dimensional space to cluster sites based on proximity to one another in the latent ecological gradient (e.g. K-means, Macqueen, 1967).

While algorithm-based ordination and clustering techniques are easily implemented and can provide ecologists with interpretable results, they present challenges when conducting inference. For example, algorithmic projection techniques do not allow researchers to explicitly account for properties of the data, including anticipated mean–variance relationships that are common in ecological data (Hui, 2016). Additional challenges arise when clustering ordination results, as it is usually impossible to determine the number of clusters present in the latent ecological gradient prior to analysis, yet this value is required for most algorithm-based clustering techniques. Finally, the algorithm-based framework does not provide a likelihood. The lack of a likelihood precludes probabilistic assessments of

distance measures, projection methods, clustering techniques or the number of clusters imposed in the latent space.

Recently, model-based approaches for unconstrained ordination have been explored that allow for measures of uncertainty on parameter estimates (Hoegh & Roberts, 2020; Hui et al., 2015; van der Veen et al., 2021). These techniques use latent variables to represent sample locations along an underlying ecological gradient. Hui et al. (2015) described a number of advantages of model-based ordination, including: (1) the ability to account for statistical properties of the data; (2) access to model assessments via residual analyses; and (3) the ability to compare models formally and quantify uncertainty in findings. The first of these three advantages allows for appropriate modelling of mean–variance relationships that are common in ecological data; failing to account for these relationships can result in ordination results that confound trends in space with changes in dispersion, resulting in misleading ordination results (Warton et al., 2012). The second and third advantages are a result of representing the data with a probabilistic model, affording likelihood-based comparisons and uncertainty on parameter estimates. While these model-based ordination techniques provide many advantages over the algorithmic ordination techniques, they still do not provide a model-based approach to cluster sample locations in the latent ecological gradient or any way to determine the number of clusters present in that space.

Following Hui et al. (2015), Hui (2016) developed a hierarchical latent variable model that allows for simultaneous clustering and ordination of ecological abundance data, known as the CORAL model, by drawing the latent variables representing the positions of sample locations in the latent ecological gradient from a finite mixture distribution. Through this process, the authors leveraged the joint information available for dimension reduction and clustering, and in doing so, outperformed competing algorithm-based clustering and ordination strategies. Additionally, the model allows for probabilistic statements about cluster membership across sites, allowing ecologists to quantify impacts of management actions with appropriate uncertainty on cluster assignments. However, the CORAL model does not afford probabilistic statements about the *number of clusters* present in the latent ecological gradient, as the latent coordinates of each site along the underlying gradient are drawn from a finite mixture model, requiring prior specification of the number of groups present. Instead, prior information and/or model selection tools (such as information criterion) are required to choose the number of clusters present, prohibiting direct probabilistic statements about the number of clusters in the latent space.

Motivated by two empirical datasets, we propose an ordination model, hereafter the Dirichlet process ordination model or DPORD, that simultaneously performs clustering and unconstrained ordination of species assemblage data and is capable of estimating the number of groups that are present in the latent ecological gradient with appropriate measures of uncertainty. This model draws the coordinates of each sample location in the latent ecological gradient from a Dirichlet process mixture model (Ferguson, 1973; Li et al., 2019; Neal, 2000; Rasmussen, 1999), thereby allowing the data to directly

inform the number of clusters in the latent space, resulting in probabilistic statements about the number of groups present in the latent space. The DPORD model is fitted to two empirical datasets using Markov chain Monte Carlo (MCMC) methods, and data and code are provided in the appendix. For comparative purposes, the first dataset considered is the Doubs river dataset analysed by Hui (2016); this dataset describes detection or non-detection of 27 species of fish along the Doubs river in eastern France. The second dataset considered describes detection or non-detection of 25 plant species collected from sagebrush steppe ecosystems in Craters of the Moon National Monument and Preserve (CRMO) in Idaho, USA in 2019. Application of the DPORD model to the Doubs river data resulted in identification of two distinct ecological communities corresponding to upstream and downstream samples locations, consistent with previous analyses. Application of the DPORD model to the CRMO data resulted in identification of three ecological communities, corresponding to a number of different ecological gradients within the monument and preserve, including elevation and proximity to active volcanic features.

2 | METHODS

2.1 | Dirichlet process mixture models

The Dirichlet process (DP; Antoniak, 1974; Ferguson, 1973) is a stochastic process used in Bayesian nonparametric models of data, including Dirichlet process mixture models (DPMM; Escobar & West, 1995; Neal, 2000; Rasmussen, 1999; Teh, 2010). The Dirichlet process is a distribution over distributions, and realisations from a DP are discrete with probability one, providing a natural mechanism for clustering (Ferguson, 1973). The DP is characterised by a concentration parameter, δ , and base distribution, G_0 . The base distribution represents the mean of the DP, and the concentration parameter can be interpreted as the inverse of the variance; as the concentration parameter increases, the DP concentrates more of its mass around the mean (Teh, 2010). For a complete treatment of Dirichlet processes, see Ferguson (1973) and Teh (2010).

The DPMM leverages the discrete realisations of the DP to construct a countably infinite mixture model (Ferguson, 1983; Neal, 2000). We begin by introducing notation for the Dirichlet process mixture model, following Neal (2000). Let $y_1, \dots, y_n$ denote the observed data, which are assumed to be drawn independently from an unknown distribution. The distribution from which the y_i are drawn is modelled as a mixture of distributions, $F(\phi)$, with G representing the mixing distribution over ϕ , where ϕ is the vector of parameters describing F . A Dirichlet process with concentration parameter δ and base distribution G_0 is chosen as the prior distribution for G , completing the DPMM. Formally, the DPMM is written as follows:

$$\begin{aligned} y_i | \phi_i &\sim F(\phi_i), \\ \phi_i | G &\sim G, \\ G &\sim \text{DP}(\delta, G_0). \end{aligned} \quad (1)$$

Since realisations of the Dirichlet process are discrete, the DP prior on G (Equation 1) generates discrete realisations of the latent-class-specific parameters, ϕ_i . Therefore, a finite number of distinct values are generated for the latent-class-specific parameters, and each ϕ_i assumes one of those values, resulting in natural clustering of the response, y_i , as each response is associated with a set of latent-class-specific parameters that assumes one of a finite number of discrete values. To further elucidate the Dirichlet process as a clustering mechanism, it is helpful to introduce a cluster label for each observation, c_i , where c_i assumes an integer value denoting to which cluster observation i belongs. Then, the discrete realisations of the Dirichlet process generate latent-class-specific parameters such that $y_i | c_i \sim F(\phi_{c_i})$, where ϕ_{c_i} denotes the set of latent-class-specific parameters associated with cluster c_i , thus making the clustering apparent. The number of distinct values assumed by the ϕ_i , or equivalently c_i , provides a data-driven estimate of the total number of clusters present, K (Li et al., 2019; Neal, 2000).

The DPMM has been used extensively in ecological applications for a variety of purposes, including: mixed-membership clustering in animal movement and abundance models (Cullen et al., 2022; Valle et al., 2017, 2018), nonparametric modelling of animal abundance (Dorazio et al., 2008; Dunson & Rodríguez, 2011), clustering of trends in animal abundance (Johnson et al., 2013), determination of population structure via genetic data (Huelsenbeck & Andolfatto, 2007), and for clustering of species behaviour and dimension reduction in joint species distribution models (Taylor-Rodríguez et al., 2017). Notably, Valle et al. (2022) discussed the ubiquity of clustering problems in ecological applications and highlights the utility of DPMMs as a clustering mechanism, capable of both estimating model parameters and the number of clusters present. In Section 2.2, we propose a Dirichlet process ordination model (DPORD) that builds on this literature by leveraging a DPMM to cluster sample locations based on species composition in the framework of unconstrained ordination.

2.2 | Dirichlet process ordination model

The DPORD model can be viewed as building upon the CORAL model proposed by Hui (2016), replacing the finite mixture model with an infinite mixture model (Equation 1). Both models are designed to fit multivariate community data, which are typically represented by an $n \times s$ community matrix, where n represents the total number of sample locations (sites) and s represents the number of observed species across all sample locations. Entries in the community matrix represent measures of species abundance, typically as binary presence-absence responses (truncated abundance), counts or ordinal classes.

Similar to Hui (2016), the DPORD model represents abundance data with a two-stage hierarchy. The first stage of the hierarchy relates the mean response to d latent variables through a generalised linear model framework. Let $i = 1, \dots, n$ index the site and $j = 1, \dots, s$ index the species. Then,

$$y_{ij} | \mu_{ij} \sim f(y_{ij} | \mu_{ij}), \quad g(\mu_{ij}) = \alpha_i + \beta_j + \mathbf{z}_i^T \boldsymbol{\theta}_j, \quad (2)$$

where y_{ij} is the observed response for species j from site i , $f(\cdot)$ represents the assumed distribution for response y_{ij} , μ_{ij} represents the mean response for species j from site i , and $g(\cdot)$ represents an appropriate link function. Within the link function, α_i represents a fixed site effect that accounts for overall differences in abundance across sites, β_j represents a fixed species effect that accounts for overall differences across species, $\mathbf{z}_i$ represents the $d \times 1$ column-vector of latent coordinates describing the location of site i in the latent ecological gradient, and $\boldsymbol{\theta}_i$ represents the $d \times 1$ column-vector of species-specific coefficients, where d is the dimension of the ordination.

The second stage of the hierarchy draws the latent vectors from a DPMM of multivariate normal distributions. Assuming the notation of Neal (2000), substituting $\boldsymbol{\mu}_i$ for the general ϕ_i described in Equation 1, the DPMM is described as follows:

$$\begin{aligned} \mathbf{z}_i | \boldsymbol{\mu}_i, \boldsymbol{\Sigma} &\sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}), \\ \boldsymbol{\mu}_i | G &\sim G, \\ G &\sim \text{DP}(\delta, G_0), \\ G_0 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \end{aligned} \quad (3)$$

where $\boldsymbol{\mu}_i$ denotes the mean vector associated with site i , $\boldsymbol{\Sigma}$ represents the variance-covariance matrix, δ represents the concentration parameter for the Dirichlet process, and G_0 represents the base distribution. In the context of ordination, δ affects the prior probability of forming new clusters in the latent ecological gradient and G_0 is typically chosen to be a multivariate normal distribution with mean vector $\boldsymbol{\mu}_0$ and covariance matrix $\boldsymbol{\Sigma}_0$, as reflected in Equation 3.

Similar to other latent variable techniques for ordination, the latent vectors $\mathbf{z}_i$ can be thought of as coordinates describing each sample location's position along the d -dimensional underlying ecological gradient. If the site-specific intercepts, α_i , are omitted, this space represents relative species abundance; if the site-specific intercepts are included, the model is adjusted for site-specific abundance and this space represents species composition (Hui, 2016). Posterior mean estimates of the latent vectors associated with each sample location may be plotted to visualise patterns in species composition or relative abundance across sample locations. By drawing these latent vectors from an infinite mixture of normal distributions, sample locations are clustered into groups of similar composition based on proximity to one another in the underlying ecological gradient. Additionally, the use of an infinite mixture allows the data to inform the number of clusters present in the ecological gradient, rather than specifying that quantity a priori.

2.2.1 | Identifiability constraints and priors

In order to estimate the DPORD model described in Equations 2 and 3 using Markov Chain Monte Carlo (MCMC) methods, identifiability constraints must be imposed and the label-switching problem must be resolved; see Appendix A for a complete description of the identifiability constraints imposed and the label-switching algorithm

implemented. Additionally, prior distributions must be specified for the site-specific intercepts α_i , species-specific intercepts β_j , species-specific coefficients $\boldsymbol{\theta}_j$, concentration parameter of the Dirichlet process δ , and the mean of the base distribution of the Dirichlet process $\boldsymbol{\mu}_0$. Since both analyses provided in Section 3 concern presence/absence data, identical prior distributions are imposed for each analysis. Independent standard normal prior distributions were placed on all site-specific intercepts, species-specific intercepts and unconstrained elements of the species-specific coefficients matrix; half-standard-normal distributions were placed on the diagonal elements of the species-specific coefficients matrix. A standard multivariate normal prior distribution was placed on the mean of the base distribution of the Dirichlet process. Unit-variance normal priors were selected for the coefficients in the DPORD model for these analyses because they resulted in weakly informative prior distributions on the probability scale, but also helped to regularise the posterior distributions by ruling out unreasonable values of the probability parameters and improve model convergence (Stan Development Team, 2023). More diffuse prior distributions may be specified if the data describe abundance, rather than species composition. A Gamma hyper-prior distribution with shape parameter of one and rate parameter of 15 was placed on the Dirichlet process concentration parameter; choosing the rate parameter to be large relative to the shape parameter encourages clustering in Dirichlet process mixture models (Frühwirth-Schnatter & Malsiner-Walli, 2018). This particular prior distribution, recommended by Frühwirth-Schnatter and Malsiner-Walli (2018), induces sparsity in the clustering results and has been shown to prevent the Dirichlet process from overfitting the underlying mixture. In Section 2.2.2, we compare this sparsity inducing prior to the common Gamma(2, 4) prior suggested by Escobar and West (1995) in the context of ordination via simulation.

2.2.2 | Comparison to other methods

The novelty of the DPORD model comes from its use of an infinite mixture model in the latent space, allowing for data-driven estimation of the number of clusters present in that space. Recently, sparse finite mixture models have been introduced as an alternative tool to estimate the number of clusters present in a mixture model (Frühwirth-Schnatter & Malsiner-Walli, 2018; Malsiner-Walli et al., 2016). These models rely on deliberately overfitting a finite mixture model with sparsity inducing prior distributions on the component weights, allowing for estimation of the number of non-empty components. However, Frühwirth-Schnatter and Malsiner-Walli (2018) showed that nearly equivalent clustering results may be obtained by matching prior distributions on the parameters that determine the weight distribution in sparse finite mixture models and Dirichlet process mixture models. Additionally, Frühwirth-Schnatter and Malsiner-Walli (2018) noted that the choice of prior distribution on the parameters that determine the weight distribution is far more influential on the clustering results than is whether a sparse finite mixture or Dirichlet process mixture is considered. Using either

Dirichlet process mixture models or sparse finite mixture models, practitioners are able to produce ordination and clustering results based on the most likely number of clusters present in the latent space a posteriori based on a single fitted model. Conversely, existing methods for ordination, including two-step algorithmic approaches and model-based approaches such as CORAL, rely on a combination of ecological expertise and model-selection tools to choose the appropriate number of clusters in the latent space.

Relying on estimation of the number of latent clusters streamlines the analytical process for practitioners by avoiding some of the difficulties that often accompany multimodel inference and model selection via information criterion (Anderson & Burnham, 2002; Grueber et al., 2011). Notably, determination of the number of latent clusters via information criterion may require fitting a large set of candidate models for comparison; for large datasets with many potential clusters, this process can become computationally burdensome for practitioners, as each model needs to be estimated and assessed prior to the model comparison. Additionally, there are numerous criterion to choose from with various penalty terms (Gelman et al., 2013), further complicating the multimodel inference that is required when implementing the two-step and finite-mixture based approaches. Rasmussen (1999) summarise these points, noting the following advantages of infinite mixture models over their finite counterparts: (1) the number of represented classes is automatically determined, (2) in many applications, it may be more appropriate not to limit the number of classes, and (3) it is much simpler to handle infinite mixture models than to work with finite models with unknown sizes.

In addition to these qualitative benefits, we provide a quantitative comparison of the Dirichlet process, sparse finite mixtures, the CORAL model, and non-metric multidimensional scaling (NMDS, Kruskal, 1964) with PAM clustering as tools for ordination and clustering via simulation in Appendix B. In this simulation, we compare the ability of each of these methods to produce an ordination, determine the number of latent groups, and recover pairwise cluster membership between sample locations. Additionally, we consider multiple prior structures for the concentration parameter

in the DPORD model, including a Gamma(2, 4) prior as suggested by Escobar and West (1995) and a Gamma(1, 15) meant to induce sparsity, as suggested by Frühwirth-Schnatter and Malsiner-Walli (2018). The results of the simulation study indicate that the model-based approaches produce comparable ordination results to each other and outperform the two-step algorithmic approach, as all model-based approaches resulted in similar Procrustes error between the estimated ordination and true generating coordinates (Legendre & Legendre, 2012), which was much lower than the error associated with the two-step approach. The DPORD model outperformed all other methods as a clustering mechanism, both resulting in the greatest proportions of correct pairwise cluster membership and correctly recovering the number of latent groups most frequently among all methods considered. The two priors considered for the DPORD model performed similarly, with the sparsity inducing prior resulting in greater certainty in the estimated number of latent groups. Therefore, we consider the DPORD model with sparsity inducing priors for both analyses provided in Section 3.

3 | DATA ANALYSES

3.1 | Doubs river data analysis

The first dataset considered describes detection or non-detection of $s = 27$ species of fish at $n = 30$ sample locations along the Doubs river in eastern France; these data are available in the ade4 R package (Dray & Dufour, 2007). Previous analysis of the Doubs river data with the CORAL model found evidence of two distinct ecological communities, partly corresponding to spatial separation in upstream and downstream sample locations (Hui, 2016). These data are again analysed with the DPORD model to allow comparison and assess model fit when the number of clusters in the latent space is assumed known a priori. Following Hui (2016), site eight was removed prior to analysis, as no species were recorded at that sample location resulting in no information regarding species composition. A map of sample locations is provided in Figure 1.

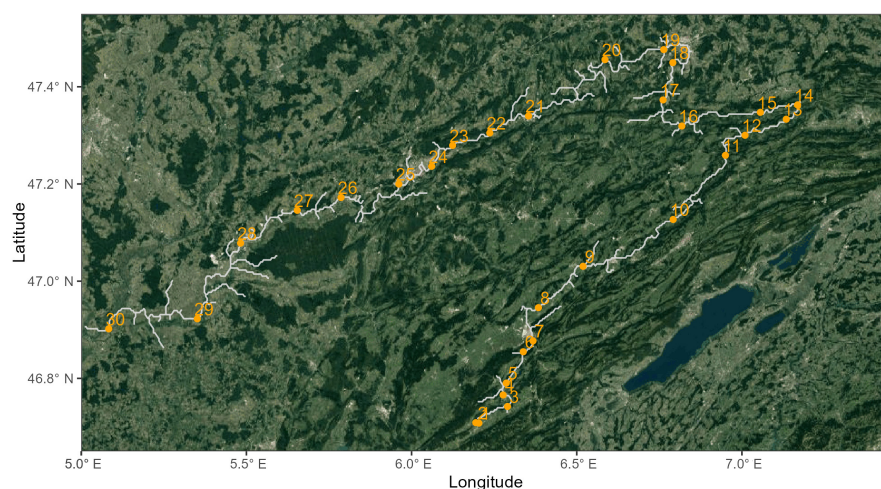


FIGURE 1 Map of 30 sample locations along the Doubs river and smaller tributaries (grey) in eastern France. Background mapping layers provided by Google mapping services.

The DPORD model was fit to the Doubs river data using the probabilistic programming language JAGS (Plummer, 2023); annotated code is provided in Appendix A. As is often the case when conducting ordination, the DPORD model was fit with two dimensions in the latent space to facilitate visualisation of the results. Four independent MCMC chains with random starting values were run for 100,000 iterations and the first 50,000 iterations of each chain were discarded as warm-up; the MCMC routine was assessed for convergence both visually and through the Gelman-Rubin statistic (Brooks & Gelman, 1998), which was calculated after adjusting the MCMC for label-switching.

The posterior mode for K , the number of clusters in the latent ecological gradient, for the Doubs river data was two (posterior probability of 0.70), with posterior probabilities of 0.26 and 0.04 for three and four or more clusters, respectively. The DPORD modelling framework is advantageous as it allows for probabilistic statements about the number of clusters in the latent ecological gradient, rather than relying on information criterion to select the appropriate number of clusters. For this analysis, the most likely number of clusters

in the latent space is two with a high degree of certainty. Thus, we focus on the case of $K = 2$ clusters in the discussion below. See Appendix C for additional analyses investigating the sensitivity of this analysis to the priors specified for the Dirichlet process concentration parameter.

The top panel of Figure 2 depicts a biplot and a map of sample locations coloured by posterior modal cluster assignments. The biplot represents the posterior mean estimates of the latent coordinates for each sample location coloured by the posterior modal cluster assignment and includes vectors representing factor loadings for seven commonly observed species: *Alburnus alburnus* (ALAL), *Gobio gobio* (GOGO), *Leuciscus cephalus* (LECE), *Nemacheilus barbatulus* (NEBA), *Phoxinus phoxinus* (PHPH), *Rutilus rutilus* (RURU) and *Thymallus thymallus* (THTH). The two distinct clusters in Figure 2 roughly correspond to upstream and downstream sample locations, which reflects previous analyses of the Doubs river data (Hui, 2016). Posterior probabilities of cluster assignments for each sample location are provided in the top panel of Figure 3. This figure provides further support for the separation in upstream and downstream

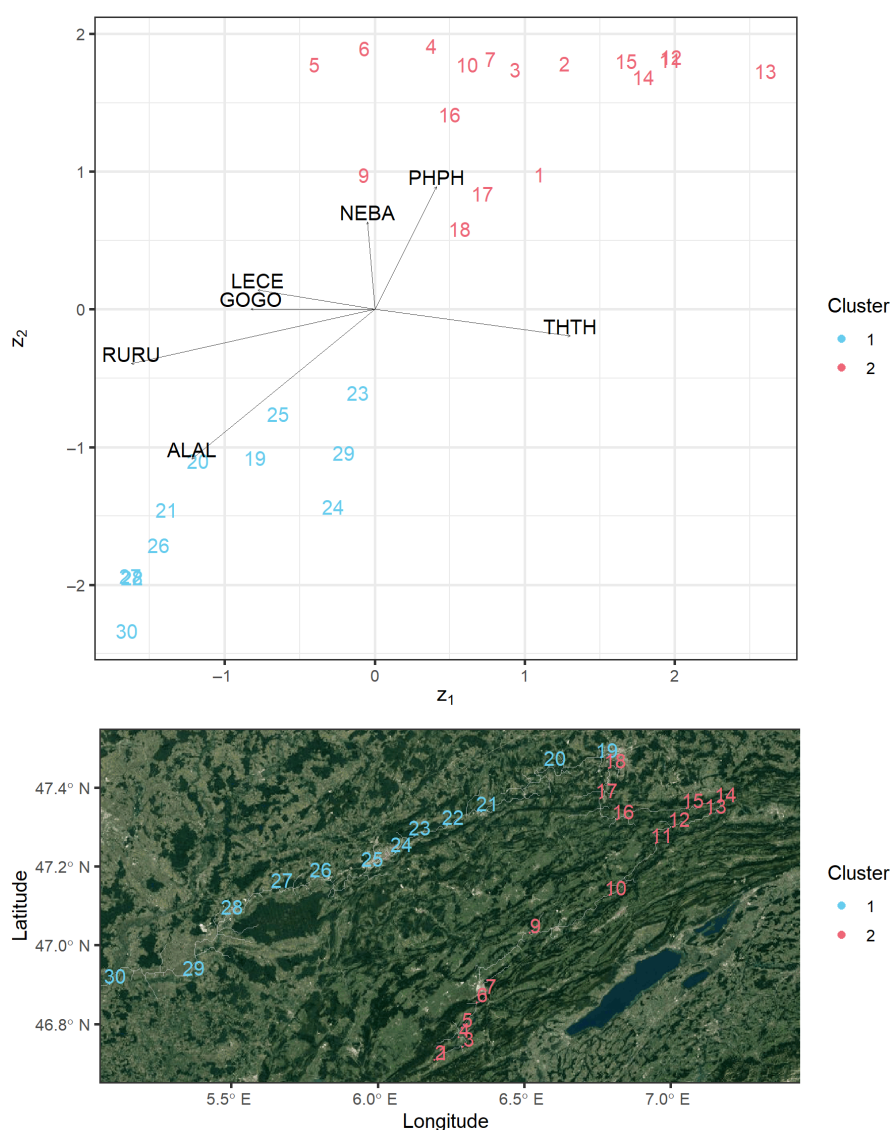


FIGURE 2 (Top) Posterior mean latent coordinates of each sample location in the Doubs river data coloured by the posterior modal cluster assignment. Vectors represent factor loadings associated with a subset of commonly observed species: *Alburnus alburnus* (ALAL), *Rutilus rutilus* (RURU), *Gobio gobio* (GOGO), *Leuciscus cephalus* (LECE), *Nemacheilus barbatulus* (NEBA), *Phoxinus phoxinus* (PHPH) and *Thymallus thymallus* (THTH). (Bottom) Sample locations coloured by posterior modal cluster assignment. The two clusters in the latent space roughly correspond to upstream and downstream sample locations, with the red cluster representing upstream sites and the blue cluster representing downstream sites. Background mapping layers provided by Google mapping services.

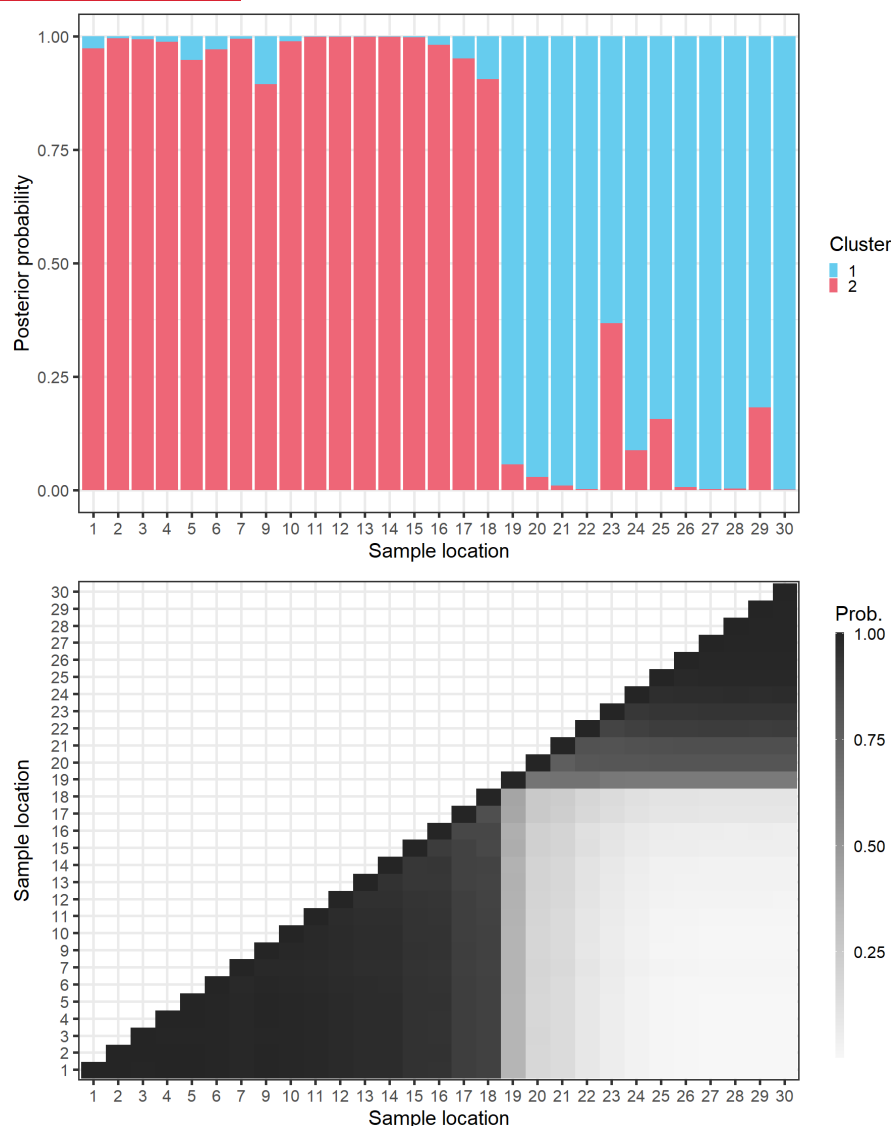


FIGURE 3 (Top) Posterior probability of cluster assignment by sample location. Sample locations one through 16 and 17 through 30 represent downstream and upstream locations, respectively. Sample locations 16, 17 and 18 are geographically located on the transition between upstream and downstream locations, potentially explaining their lower modal cluster probabilities. (Bottom) Pairwise, posterior cluster membership probabilities between sample locations in the Doubs river data. Each cell in this plot represents the posterior probability that the sample locations on the horizontal and vertical axis share a cluster in the latent space.

locations. Sample locations 16, 17 and 18, which are geographically located in the transition from upstream to downstream, have low probabilities of inclusion relative to other sample locations in their respective clusters.

In addition to posterior modal cluster assignments, it can be informative to look at pairwise, posterior cluster membership between sample locations (bottom panel, Figure 3). Each cell in this figure represents the posterior probability that the sample location on the horizontal and vertical axis share a cluster in the latent space. This figure further elucidates the separation between upstream and downstream sample locations but also describes the degree of certainty in pairwise cluster membership. For example, sample location 19, which shares a posterior modal group assignment with locations 20 through 30, has a moderate pairwise

cluster membership probability with each of locations 1 through 18, suggesting some degree of similarity between location 19 and locations 1 through 18.

In addition to their geographic location, other factors contribute to the clustering results. Notably, the upstream locations (cluster two, coloured in red) are comprised of a diverse collection of species; 22 of the 27 species were observed in at least half of the sample locations assigned to cluster two, and ALAL, RURU and LECE were observed in all sample locations assigned to cluster two. Conversely, the downstream locations (cluster one, coloured in blue) were considerably less diverse, with only five species present in at least half of the sample locations: GOGO (present in 53% of sample locations), LECE (present in 76% of sample locations), SATR (present in 94% of sample locations), PHPH (present in 94%

of sample locations) and NEBA (present in 94% of sample locations). Notably, only three species were recorded at sample location 23 (LECE, RURU and ALAL), resulting in a considerably less diverse collection of species when compared to other upstream locations, perhaps explaining its large probability of being clustered with the downstream sample locations despite its upstream location, relatively to other upstream locations.

3.2 | Craters of the Moon National Monument and Preserve data analyses

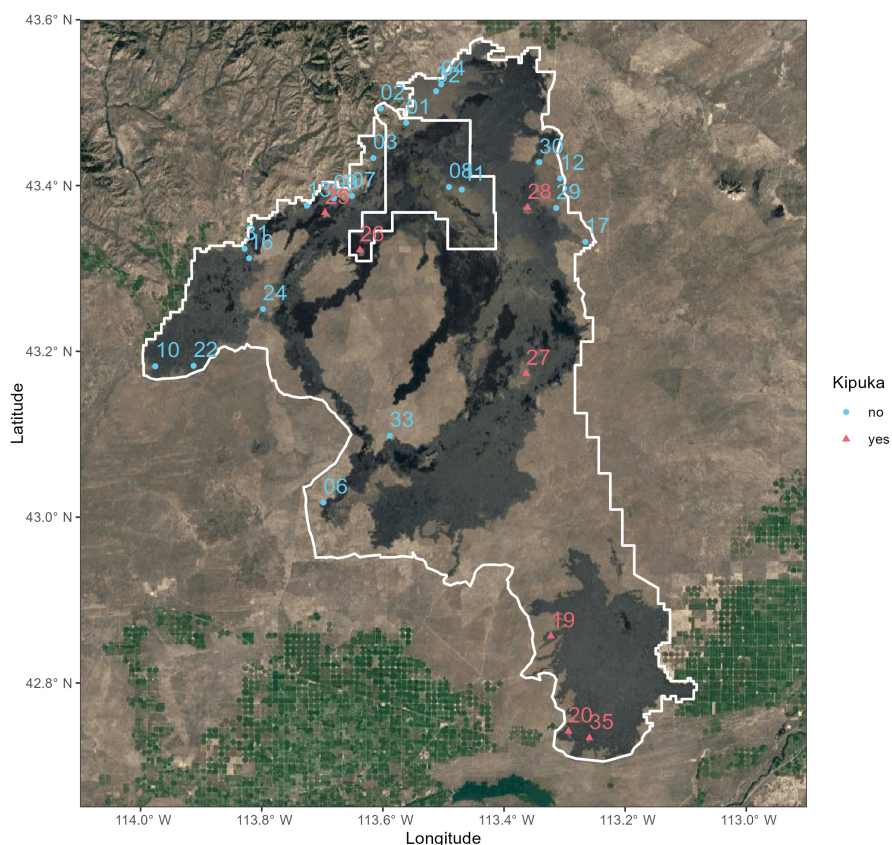
The second dataset considered describes presence or absence of 78 plant species across 28 approximately 10 hectare sample frames in the Craters of the Moon National Monument and Preserve in Idaho, USA (CRMO). Craters of the Moon is located in a sagebrush steppe ecosystem, which covers much of the Great Basin and Snake River Plain that extends across the Interior West of the United States. Vegetation monitoring in CRMO presents a unique opportunity to explore ordination and clustering methods, as the park is comprised of a combination of volcanic features and sagebrush steppe along an elevational gradient, leading to anticipated differences in plant composition based on proximity to volcanic features and elevational gradient position.

Sample frame locations were chosen to represent the range of community types within the CRMO sagebrush steppe landscape and were arranged along the proximity to volcanic features and elevation

gradients (Yeo et al., 2009). Within each delineated sample frame, approximately 50 one square meter quadrat locations were selected according to the Generalised Random Tessellation Stratified (GRTS) spatially balanced sampling design (Stevens Jr. & Olsen, 2004). A total of 1518 vegetation quadrat locations were included in our analysis. For each quadrat location, ordinal cover classes for all plant species present in the one square meter quadrat were recorded. These ordinal classes were converted to presence-absence records prior to analysis. Sample frame locations are provided in Figure 4. The primary analytical goal for the CRMO data was to assess the similarity in species composition across the $n = 28$ sample frames in order to better understand community types within the monitoring program and to inform management strategies for preserving the sagebrush communities. Sample frames 19, 20, 25, 26, 27, 28 and 35 were of particular interest, as they are located in kipukas. Kipukas are areas of deeper soiled vegetation within the lava fields in CRMO that are completely surrounded by younger lava flows. Consequently, these sample frames have been physically isolated from livestock grazing and represent potentially quasi-pristine plant communities, making them of particular conservation interest (Nicolli, 2019).

Prior to analysis, all species that did not appear in at least five percent of sample locations were excluded, resulting in $s = 25$ species considered for analysis. Removal of rare species prior to ordination is common in many practical settings, as common species are often most important for assessing relationships between species composition or abundance and underlying environmental gradients (Brasil et al., 2020). To account for the nested sampling design,

FIGURE 4 Satellite map of 28 sample frame locations in Craters of the Moon National Monument and Preserve in Idaho, USA. The interior polygon defines the National Monument and the exterior polygon defines the National Preserve. Sample frames 19, 20, 25, 26, 27, 28 and 35 are located in unique vegetated islands within lava flows called kipukas (coloured in red), and are of particular conservation interest, representing potentially pristine communities not physically disturbed by grazing or other anthropogenic activities (Yeo et al., 2009). Background satellite imagery layers provided by Google mapping services.



presence-absence records at the quadrat-level were aggregated to form Binomial counts for each sample frame. The DPORD model was fit with $d = 2$ dimensions in the latent space to allow for visualisation and interpretation of the resulting ordination and clustering using JAGS. Four independent MCMC chains with random starting values were run for 200,000 iterations, and the first 100,000 iterations were discarded as warm-up.

The posterior modal number of clusters in the latent space for the CRMO data was three (posterior probability of 0.42, with posterior probabilities of 0.05, 0.35, 0.15 and 0.03 for one, two, four and five or more clusters, respectively.) Below, we focus discussion on the case where $K = 3$, as it is the most likely number of clusters in the latent space. Figure 5 depicts a biplot and map of sample locations coloured by posterior modal cluster assignment. The biplot includes vectors representing factor loadings for a subset of observed species: *Allium* spp. (ALLSPP), *Artemisia tridentata* (ARTTRID), *Crepis acuminata* (CREACU), *Elymus elymoides* (ELYELY), *Ericameria nana* (ERINAN), *Penstemon* spp. (PENSPP) and *Poa secunda* (POASEC). Posterior probabilities of cluster assignments are provided in the

top panel of Figure 6. Clustering results largely aligned with ecological gradients that exist within the park. For example, cluster two, comprised of samples frames 1, 2, 8, 11, 25 and 28, contains many of the high elevation sample locations, including four of the five highest elevation locations in the park. These four sample frames are located along the highest ridge in CRMO, which spans from the Pioneer mountains north of the monument down to the visitor centre. The sample frames in cluster two were largely dominated by big sagebrush (*Artemisia tridentata*, present in 58% of quadrat locations) and wild buckwheat (*Eriogonum* spp., present in 56% of quadrat locations), though wild bluegrass (*Poa secunda*) and cheatgrass (*Bromus tectorum*) were common as well, present in 50% and 48% of quadrat locations, respectively.

Clustering results also corroborated anticipated differences in species composition between the sample frames located in kipukas and those elsewhere in the park. Cluster one (frames 19, 20, 26, 27, 32 and 35) accounts for five of the seven sample frames located in kipukas, with the remaining two frames included in cluster two. Quadrat locations from sample frames in cluster one were largely dominated

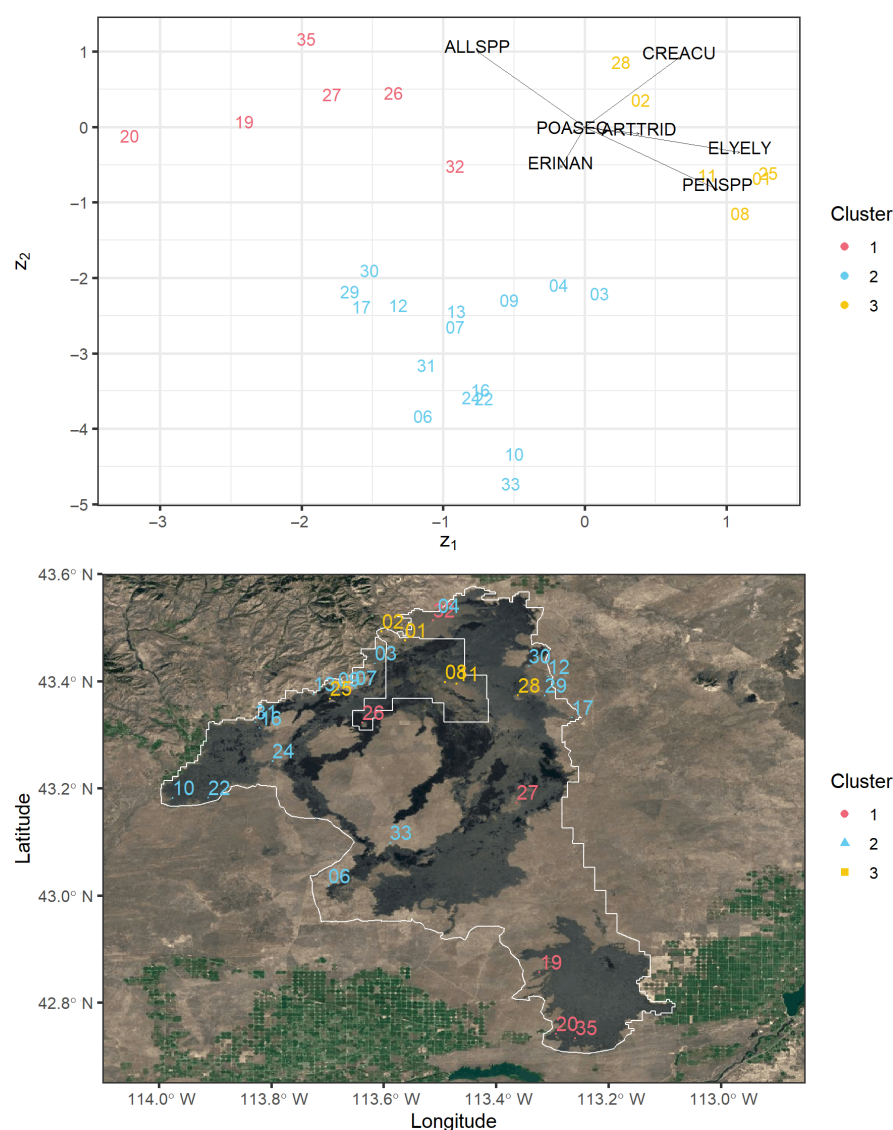


FIGURE 5 (Top) Posterior mean latent coordinates of each sample location in CRMO coloured by the posterior modal cluster assignment. Vectors represent factor loadings associated with a subset of commonly observed species: *Allium* spp. (ALLSPP), *Artemisia tridentata* (ARTTRID), *Crepis acuminata* (CREACU), *Elymus elymoides* (ELYELY), *Ericameria nana* (ERINAN), *Penstemon* spp. (PENSPP) and *Poa secunda* (POASEC). (Bottom) Sample locations coloured by posterior modal cluster assignment. Cluster one is denoted in red and is comprised of sample frames 19, 20, 26, 27, 32 and 35; cluster two is denoted in blue and is comprised of sample frames 1, 2, 8, 11, 25 and 28; all other sample frames are in cluster three, denoted in gold. Background mapping layers provided by Google mapping services.

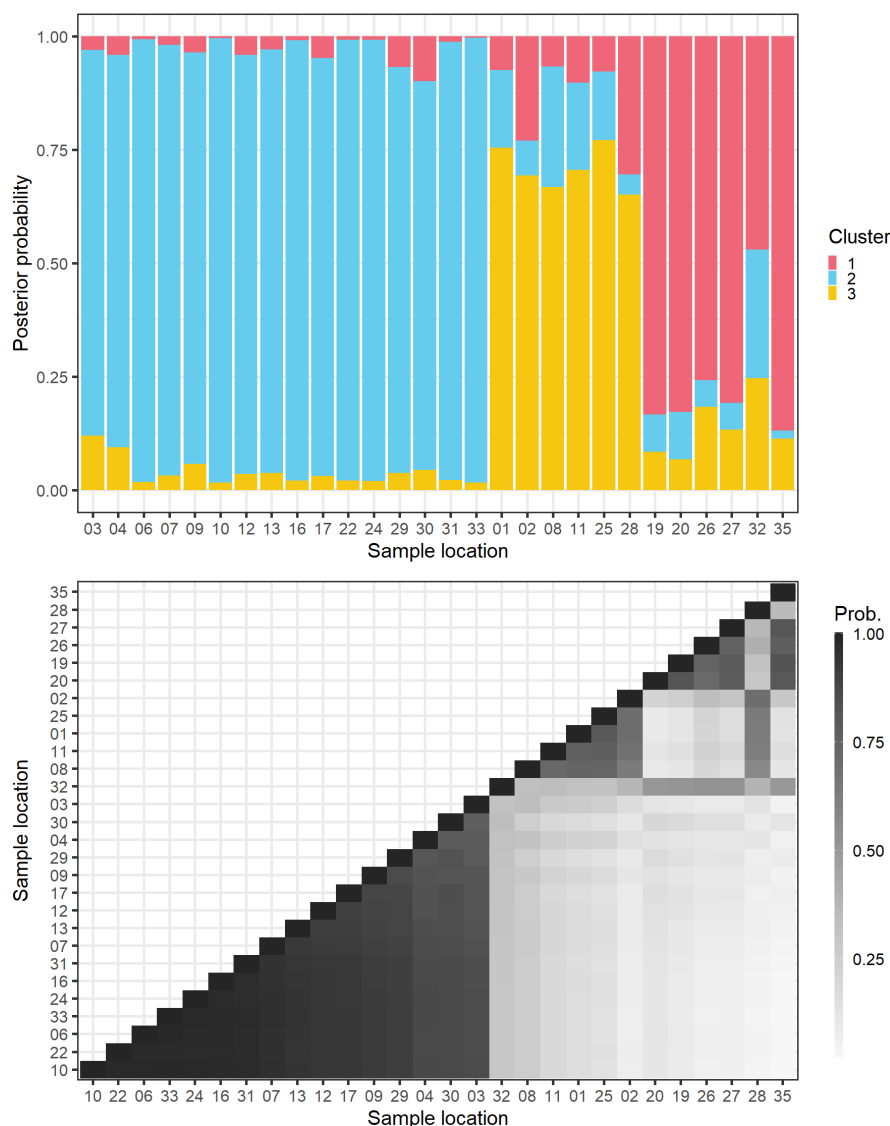


FIGURE 6 (Top) Posterior probability of cluster assignment by sample frame. (Bottom) Pairwise, posterior cluster membership probabilities between sample frames. Each cell in this plot represents the posterior probability that the sample locations on the horizontal and vertical axis share a cluster in the latent space.

by cheatgrass (present in 99% of quadrat locations) and wild bluegrass (present in 66% of quadrat locations) but tumbleweed mustard (*Sisymbrium altissimum*) and *Allium* spp. were also fairly common, present in 49% and 50% of quadrat locations, respectively. Our discovery of heavy weed infestation (both cheatgrass and tumbleweed mustard) in the cluster of kipuka frames within cluster one was surprising and suggests a process of degradation has occurred in spite of physical barriers to grazing and other land use activities. The two kipuka frames (25 and 28) clustered in group two are quite distinctive with low amounts of big sagebrush but relatively high abundance of two other native sagebrush shrubs (*Artemisia arbuscula* and *Artemisia tripartita*), and they also remain relatively uninfested by cheatgrass and other weeds. These frames, a subset of cluster two, were dominated mostly by wild bluegrass (present in 76% of quadrat locations) and needlegrass (*Achnatherum* spp., present in 68% of quadrat locations).

Additional relationships may be explored with the pairwise cluster membership plot in the bottom panel of Figure 6. Each cell in Figure 6 represents the posterior probability the sample frame on the horizontal and vertical axis share a cluster in the latent space. This figure may be used to identify similarity in sample frames not described by the posterior modal cluster assignments. For this analysis, sample frames only shared meaningful pairwise cluster membership probabilities with other frames within their cluster, suggesting clear separation between the three estimated clusters.

4 | DISCUSSION

We developed a statistical framework for simultaneous clustering and ordination of species composition data that allows for

estimation of the number of clusters present in the latent ecological gradient. Code is provided for fitting and summarising the DPORD model, facilitating easier uptake and application among ecologists and wildlife researchers. Existing methods for simultaneous clustering and ordination of abundance data rely on expert knowledge, information criterion or other model-selection tools to choose the appropriate number of clusters present in the latent ecological gradient; in these cases, inference then proceeds conditional on the number of clusters specified in a two-step process (Hui, 2016). Conversely, by incorporating Dirichlet process priors on mixture-component-level distributions in an ordination model, the DPORD model allows for data-driven estimation of the appropriate number of clusters in the latent space in a single step. As a result, ecologists and wildlife managers are afforded probabilistic statements about the number of clusters present in the latent ecological gradient and inference may proceed based on the most likely number of clusters. In Appendix B, we demonstrated via simulation that the proposed DPORD framework outperforms existing techniques for ordination and clustering, resulting in correct identification of the number of clusters in the latent space most frequently of the methods considered.

Using the DPORD model, we provided ordination results for the historical Doubs river fish dataset. Results suggested strong evidence of two distinct regions in the latent ecological gradient corresponding to upstream and downstream locations, consistent with previous analyses of these data. The DPORD model was also applied to plant cover data from Craters of the Moon National Monument and Preserve in Idaho, USA. Results suggested there exist three distinct ecological regions in the latent ecological gradient corresponding to various features of the ecological gradient in CRMO, including elevation, proximity to volcanic features and degree of ecological degradation from weed invasion. Furthermore, clustering results were consistent with anticipated differences in species composition between sample frames located in kipukas and elsewhere in the park. These clustering results provide insight into differences between various plant communities in CRMO, supporting more reliable and cost effective monitoring of sagebrush steppe throughout the park.

Considering extensions to the DPORD model for longitudinal datasets presents an interesting opportunity for future research, as community monitoring programs are often implemented over multiple years. However, hard clustering approaches like finite and infinite mixture models have been shown to have limited ability to characterise transitions in space and time when compared to soft clustering approaches, such as latent Dirichlet allocation (Blei et al., 2002) or archetypal analysis (Cutler & Breiman, 1994). In the context of model based ordination, soft clustering approaches would allow each sample location to be clustered to multiple groups, resulting in a mixed membership clustering. Mixed membership clustering is especially useful when sample locations occur, spatially or temporally, at transitions between anticipated clusters (i.e. at the transition between upstream and downstream sample locations or between periods in time). Extending the DPORD framework to better cluster transitions

in time and space through mixed membership clustering methods provides an interesting path for future research.

In this article, we have emphasised implementation of the DPORD framework via the probabilistic programming language JAGS to improve accessibility of our modelling framework for practitioners. However, implementing the DPORD model in a probabilistic programming language presents scalability challenges. For example, it took approximately 3 min to draw 100,000 samples from the DPORD model for the Doubs river data (containing 29 rows and 27 columns) on a computer with 64 GB of RAM and an Intel i9-10980XE CPU clocked at 3.00 GHz; this runtime would be expected to increase if the number of sample locations (on which the Dirichlet process is specified) also increased. In order to implement more complex or alternative clustering mechanisms and improve the scalability of the DPORD model, custom MCMC procedures should be explored. For example, reversible jump MCMC (RJMCMC) approaches (Green, 1995) could be implemented to allow for estimation of the appropriate number of dimensions for the ordination. However, implementing RJMCMC techniques could allow the dimension of the resulting ordination to exceed three, complicating interpretation of the results. Additionally, implementing custom MCMC approaches comes at the cost of accessibility for practitioners, and implementing custom samplers for Dirichlet processes, and other mixture models techniques, can present challenges for large data applications (Hastie et al., 2013). However, scalable methods and software for Bayesian data analysis continue to be developed; for example, software packages like greta present a promising future for scalable MCMC approaches (Golding, 2019). Alternatively, variational Bayesian methods may be considered, removing the computation time associated with MCMC approaches (Blei et al., 2017).

AUTHOR CONTRIBUTIONS

Christian Stratton: Conceptualisation, formal analysis, methodology(lead), software, validation, visualisation, writing—original draft, writing—review and editing; Andrew Hoegh: Conceptualisation(lead), methodology, supervision, writing—review and editing; Thomas J. Rodhouse: Data curation, investigation, writing—review and editing; Jennifer L. Green: Methodology, writing—review and editing; Katharine M. Banner: Methodology, writing—review and editing; Kathryn M. Irvine: Methodology, writing—review and editing.

ACKNOWLEDGEMENTS

We thank Dr. Mark Greenwood for his helpful comments and discussion during writing of this manuscript. Any use of trade, firm or product names is for descriptive purposes only and does not imply endorsement by the US Government. This material is based upon work supported by the U.S. Geological Survey under Grant/Cooperative Agreement no. G23AC00118.

CONFLICT OF INTEREST STATEMENT

All authors declare there are no conflicts of interest.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/2041-210X.14389>.

DATA AVAILABILITY STATEMENT

The Doubs river data are available through the ade4 R package at <https://cran.r-project.org/web/packages/ade4/index.html> (Dray & Dufour, 2007). The CRMO data, and all code used to analyse both datasets, are publicly available and archived on Zenodo at <https://zenodo.org/records/11200992> (Stratton et al., 2024).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix A. DPOD model implementation.

Appendix B. Comparison of ordination approaches.

Appendix C. Sensitivity analyses.

How to cite this article: Stratton, C., Hoegh, A., Rodhouse, T. J., Green, J. L., Banner, K. M., & Irvine, K. M. (2024). Clustering and unconstrained ordination with Dirichlet process mixture models. *Methods in Ecology and Evolution*, 15, 1720–1732. <https://doi.org/10.1111/2041-210X.14389>